MSG No. 207.41 $P4321'$ [Type II, cubic]

Table 1: Wyckoff site: 1a, site symmetry: 4321'

No.	position	mapping
1	[0, 0, 0]	[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24,
		25,26,27,28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,44,45,46,47,48]

Table 2: Wyckoff site: 1b, site symmetry: 4321'

No.	position	mapping
1	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24,
		25,26,27,28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,44,45,46,47,48]

Table 3: Wyckoff site: 3c, site symmetry: 42.21'

No.	position	mapping
1	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	[1,2,3,8,9,10,13,14,25,26,27,32,33,34,37,38]
2	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	[4,5,15,16,18,19,21,23,28,29,39,40,42,43,45,47]
3	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	[6,7,11,12,17,20,22,24,30,31,35,36,41,44,46,48]

Table 4: Wyckoff site: 3d, site symmetry: 42.21'

No.	position	mapping
1	$\left[\frac{1}{2}, 0, 0\right]$	[1,2,3,8,9,10,13,14,25,26,27,32,33,34,37,38]
2	$\left[0, 0, \frac{1}{2}\right]$	[4,5,15,16,18,19,21,23,28,29,39,40,42,43,45,47]
3	$\left[0, \frac{1}{2}, 0\right]$	[6,7,11,12,17,20,22,24,30,31,35,36,41,44,46,48]

Table 5: Wyckoff site: 6e, site symmetry: 4..1'

No.	position	mapping
1	$[x, 0, 0]$	[1,2,3,8,25,26,27,32]
2	$[0, 0, -x]$	[4,16,19,23,28,40,43,47]
3	$[0, 0, x]$	[5,15,18,21,29,39,42,45]
4	$[0, x, 0]$	[6,11,17,24,30,35,41,48]
5	$[0, -x, 0]$	[7,12,20,22,31,36,44,46]
6	$[-x, 0, 0]$	[9,10,13,14,33,34,37,38]

Table 6: Wyckoff site: 6f, site symmetry: 4..1'

No.	position	mapping
1	$\left[x, \frac{1}{2}, \frac{1}{2}\right]$	[1,2,3,8,25,26,27,32]
2	$\left[\frac{1}{2}, \frac{1}{2}, -x\right]$	[4,16,19,23,28,40,43,47]
3	$\left[\frac{1}{2}, \frac{1}{2}, x\right]$	[5,15,18,21,29,39,42,45]

continued ...

Table 6

No.	position	mapping
4	$\left[\frac{1}{2}, x, \frac{1}{2}\right]$	[6,11,17,24,30,35,41,48]
5	$\left[\frac{1}{2}, -x, \frac{1}{2}\right]$	[7,12,20,22,31,36,44,46]
6	$\left[-x, \frac{1}{2}, \frac{1}{2}\right]$	[9,10,13,14,33,34,37,38]

Table 7: Wyckoff site: 8g, site symmetry: .3.1'

No.	position	mapping
1	$[x, x, x]$	[1,17,18,25,41,42]
2	$[x, -x, x]$	[2,7,15,26,31,39]
3	$[x, x, -x]$	[3,4,11,27,28,35]
4	$[-x, x, x]$	[5,6,13,29,30,37]
5	$[x, -x, -x]$	[8,22,23,32,46,47]
6	$[-x, x, -x]$	[9,19,24,33,43,48]
7	$[-x, -x, x]$	[10,20,21,34,44,45]
8	$[-x, -x, -x]$	[12,14,16,36,38,40]

Table 8: Wyckoff site: 12h, site symmetry: 2..1'

No.	position	mapping
1	$\left[x, \frac{1}{2}, 0\right]$	[1,8,25,32]
2	$\left[x, 0, \frac{1}{2}\right]$	[2,3,26,27]
3	$\left[0, \frac{1}{2}, -x\right]$	[4,16,28,40]
4	$\left[0, \frac{1}{2}, x\right]$	[5,15,29,39]
5	$\left[\frac{1}{2}, x, 0\right]$	[6,11,30,35]
6	$\left[\frac{1}{2}, -x, 0\right]$	[7,12,31,36]
7	$\left[-x, \frac{1}{2}, 0\right]$	[9,10,33,34]
8	$\left[-x, 0, \frac{1}{2}\right]$	[13,14,37,38]
9	$\left[0, x, \frac{1}{2}\right]$	[17,24,41,48]
10	$\left[\frac{1}{2}, 0, x\right]$	[18,21,42,45]
11	$\left[\frac{1}{2}, 0, -x\right]$	[19,23,43,47]
12	$\left[0, -x, \frac{1}{2}\right]$	[20,22,44,46]

Table 9: Wyckoff site: 12i, site symmetry: ..21'

No.	position	mapping
1	$[0, y, y]$	[1,13,25,37]
2	$[0, -y, y]$	[2,10,26,34]
3	$[0, y, -y]$	[3,9,27,33]
4	$[y, y, 0]$	[4,18,28,42]
5	$[-y, y, 0]$	[5,19,29,43]
6	$[-y, 0, y]$	[6,20,30,44]
7	$[y, 0, y]$	[7,17,31,41]
8	$[0, -y, -y]$	[8,14,32,38]

continued ...

Table 9

No.	position	mapping
9	$[y, 0, -y]$	`[11,22,35,46]`
10	$[-y, 0, -y]$	`[12,24,36,48]`
11	$[y, -y, 0]$	`[15,23,39,47]`
12	$[-y, -y, 0]$	`[16,21,40,45]`

Table 10: Wyckoff site: `12j`, site symmetry: `..21'`

No.	position	mapping
1	$\left[\frac{1}{2}, y, y\right]$	`[1,13,25,37]`
2	$\left[\frac{1}{2}, -y, y\right]$	`[2,10,26,34]`
3	$\left[\frac{1}{2}, y, -y\right]$	`[3,9,27,33]`
4	$\left[y, y, \frac{1}{2}\right]$	`[4,18,28,42]`
5	$\left[-y, y, \frac{1}{2}\right]$	`[5,19,29,43]`
6	$\left[-y, \frac{1}{2}, y\right]$	`[6,20,30,44]`
7	$\left[y, \frac{1}{2}, y\right]$	`[7,17,31,41]`
8	$\left[\frac{1}{2}, -y, -y\right]$	`[8,14,32,38]`
9	$\left[y, \frac{1}{2}, -y\right]$	`[11,22,35,46]`
10	$\left[-y, \frac{1}{2}, -y\right]$	`[12,24,36,48]`
11	$\left[y, -y, \frac{1}{2}\right]$	`[15,23,39,47]`
12	$\left[-y, -y, \frac{1}{2}\right]$	`[16,21,40,45]`

Table 11: Wyckoff site: `24k`, site symmetry: `11'`

No.	position	mapping
1	$[x, y, z]$	`[1,25]`
2	$[x, -z, y]$	`[2,26]`
3	$[x, z, -y]$	`[3,27]`
4	$[z, y, -x]$	`[4,28]`
5	$[-z, y, x]$	`[5,29]`
6	$[-y, x, z]$	`[6,30]`
7	$[y, -x, z]$	`[7,31]`
8	$[x, -y, -z]$	`[8,32]`
9	$[-x, y, -z]$	`[9,33]`
10	$[-x, -y, z]$	`[10,34]`
11	$[y, x, -z]$	`[11,35]`
12	$[-y, -x, -z]$	`[12,36]`
13	$[-x, z, y]$	`[13,37]`
14	$[-x, -z, -y]$	`[14,38]`
15	$[z, -y, x]$	`[15,39]`
16	$[-z, -y, -x]$	`[16,40]`
17	$[z, x, y]$	`[17,41]`
18	$[y, z, x]$	`[18,42]`
19	$[-y, z, -x]$	`[19,43]`
20	$[-z, -x, y]$	`[20,44]`

continued ...

Table 11

No.	position	mapping
21	$[-y, -z, x]$	`[21,45]`
22	$[z, -x, -y]$	`[22,46]`
23	$[y, -z, -x]$	`[23,47]`
24	$[-z, x, -y]$	`[24,48]`